Literature Review

UK Train Rides Analytics: Delays, Cancellations, Refunds and Revenue Forecasting (Power BI)

DEPI Graduation Project

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1. Introduction

This review looks at published work related to the project: the causes of railway delays, passenger compensation and refund behaviour in Great Britain, and the use of star schema modelling in Power BI. Its purpose is to show what is already known and where this project fits.

2. Causes of Railway Delays

Delays are usually studied by grouping them into cause categories and measuring how often each cause occurs and how long it lasts. Yang et al. (2019) analysed operational data from four Chinese high-speed railway lines, classified delays into eleven cause categories, and fitted statistical distributions to delay durations for each cause. Their work shows that different causes behave differently, so each cause should be analysed separately.

Nagy and Csiszar (2015) studied schedule deviations at stations of Hungarian Railways and focused on delays caused by random external factors, especially weather conditions. They suggest that correlations in historical delay data make delays predictable, which supports the use of historical data for forecasting.

Relevance to this project: our dataset also records a reason for each delay or cancellation (weather, signal failure, technical issue, staffing, staff shortage and traffic). We follow the same idea of analysing each cause separately and measuring its effect, but we measure it in revenue and link each cause to a responsible department.

3. Compensation and Refund Behaviour in Great Britain

In Great Britain, passengers can claim compensation for delays through the Delay Repay schemes. A survey carried out by Transport Focus for the Department for Transport found that only 37% of eligible passengers claimed for their most recent delay (Department for Transport, 2020). Claim rates were higher for longer delays and more expensive tickets. The most common reason for not claiming was that the amount was not worth the effort (51% of non-claimants), and some passengers did not know they could claim.

Rossa et al. (2024) studied the Delay Repay rules and estimated that operators repay between about 0.3% and 1.8% of their ticket revenue, depending on journey length. They note that the same rules apply to all ticket types and journey lengths, and call for more research on the scheme.

Relevance to this project: these studies show that many passengers who suffer a delay do not request their money back. This supports the idea of a group of customers who had a delayed or cancelled trip but did not request a refund. In our dataset this group is large (3,054 of 4,172 delayed or cancelled trips). The dataset does not record how these customers were handled, so we treat them as handled customers as an assumption, and we propose actions for them.

4. Star Schema Modelling in Power BI

Microsoft's guidance for Power BI recommends organising a model into dimension tables, which describe business entities and are used for filtering and grouping, and fact tables, which store measurable events and are used for summarisation (Microsoft, 2024). It advises against mixing the two types in one table. A star schema keeps the model simple and efficient, and makes DAX measures easier to write.

The same guidance discusses role-playing dimensions, such as a date table used for several dates. It describes two options: creating a separate dimension table for each role, or keeping one table with one active relationship and the others inactive, activated in measures when needed.

Relevance to this project: we use a star schema with one fact table and dimension tables for purchase type, payment, railcard, ticket class, ticket type, stations, status, refund, reason, date and time, and we keep DAX measures in a separate measures table.

5. Gap and Contribution

The studies above focus on the statistical analysis of delays, or on compensation schemes at the level of the whole rail system. We did not find work that combines, in one interactive dashboard, the following elements for a ticket-level dataset:
- delays, cancellations and refund requests by cause;
- the revenue affected by each cause;
- customers who did not request a refund;
- assumed solutions linked to the responsible departments, with an adjustable success rate;
- peak hours detected from the data itself.

This project contributes an interactive Power BI dashboard that brings these elements together.

References

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